Edexcel IAL Biology · Unit 2 · Topic 4A

The Cell Wall

Plant Cell Ultrastructure · Electron Microscopy · Cellulose Structure

4.1(i)
4.1(ii)
4.2
4.3

Total marks	25
Suggested time	30–35 minutes
Spec refs	4.1(i) · 4.1(ii) · 4.2 · 4.3
Command words	State · Describe · Explain · Identify · Compare

Instructions: Answer all questions. The number of marks available is shown in brackets at the end of each question. For multiple-choice questions, circle your chosen answer. For written questions, use the answer lines provided. The mark scheme begins on a new page at the end of this paper.

SECTION A — Multiple Choice (Questions 1–2) 2 marks

1 Which of the following is found in animal cells but *not* in plant cells? **(1)**

A Ribosomes

B Golgi apparatus

C Centrioles

D Mitochondria

2 A cell contains a double outer membrane, no internal thylakoid membrane system, and large electron-dense granules in its interior. Under a transmission electron microscope, this organelle is most likely to be: **(1)**

A Chloroplast

B Mitochondrion

C Amyloplast

D Nucleus

SECTION B — Short Answer (Questions 3–5) 8 marks

3 Name **three** structures found in a plant cell that are absent from an animal cell, and for each one, state its function. **(3)**

One mark per correct structure + function pair.

1. Structure: ________________________________ Function: ________________________________

2. Structure: ________________________________ Function: ________________________________

3. Structure: ________________________________ Function: ________________________________

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4 State **two** differences between how carbohydrate is stored in plant cells compared with animal cells. **(2)**

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5 The diagram below shows the structure of part of a plant cell wall. **(3)**

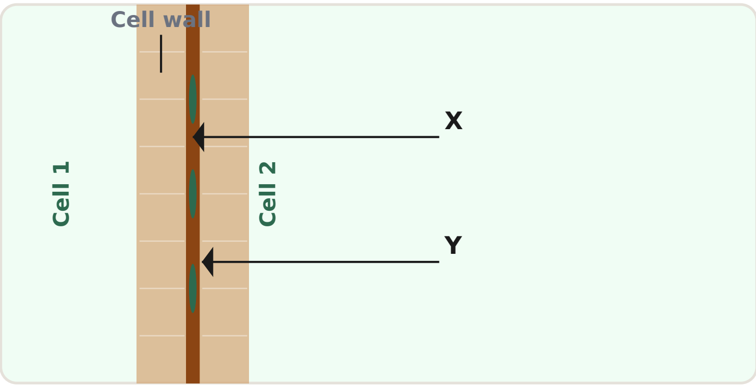

Fig. 1 — Junction between two adjacent plant cell walls (not to scale)

(a) Identify structures X and Y. **(2)**

X: ____

Y: ____

(b) State the function of structure Y. **(1)**

SECTION C — Structured Questions (Questions 6–8) 12 marks

6 A student examines a plant cell under a transmission electron microscope (TEM) and identifies two organelles: one oval with grana, and one irregular with electron-dense granules. **(4)**

(a) Name the two organelles and state **one** piece of evidence from the TEM image for each identification. **(2)**

Organelle 1 (oval, with grana)

Name: ______________________ Evidence: ______________________

Organelle 2 (irregular, electron-dense granules)

Name: ______________________ Evidence: ______________________

(b) Describe the appearance of the tonoplast in a TEM image, and state where it would be located within the plant cell. **(2)**

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7 The table below shows some features of two types of polysaccharide, P and Q, found in plant cells. **(5)**

Feature	Polysaccharide P	Polysaccharide Q
Monomer	β-glucose	α-glucose
Glycosidic bond	1,4	1,4
Chain conformation	Straight, unbranched	Helical
Location in cell	Cell wall	Amyloplast

(a) Identify polysaccharides P and Q. **(2)**

P: ______________________

Q: ______________________

(b) Use the information in the table to explain why polysaccharide P is able to form microfibrils, whereas polysaccharide Q does not. **(3)**

8 Explain how the structure of cellulose is related to its function as a structural component of the plant cell wall. *In your answer, refer to the role of β-glucose, glycosidic bonds, and hydrogen bonds.* **(6)**

This question assesses your ability to construct a coherent, sequential explanation (AO1/AO2).

SECTION D — Synoptic / Stretch (Question 9) 3 marks

9 A student states: "*The cell walls of plants and the cell walls of bacteria are both rigid structures, so they must have the same composition.*" **(3)**

Evaluate this statement.

Recall the composition of prokaryotic cell walls from Topic 3B (spec 3.5).

Total marks available for this paper **25**

MARK SCHEME 4A: The Cell Wall — All questions

*Marking guidance: Mark schemes use the following conventions where applicable: accept in italics indicates alternative acceptable wording; **bold** indicates required content for that mark point; (1) denotes one mark available. Award marks for correct science, even if not in the exact wording shown.*

Q1 *MCQ — Structure absent from plant cells* **1 mark**

• The correct answer is **C — Centrioles**. (1)

Centrioles are present in animal cells and organise the spindle during cell division. Most flowering plant cells lack centrioles.

ACCEPT: Any unambiguous indication of C.

REJECT: A, B or D.

• A incorrect: Ribosomes are found in all living cells.

• B incorrect: Golgi apparatus is found in all eukaryotes.

• D incorrect: Mitochondria are found in all eukaryotic cells including plants.

Q2 *MCQ — Organelle identification from TEM description* **1 mark**

• The correct answer is **C — Amyloplast**. (1)

An amyloplast has a double outer membrane and stores starch, which appears as very electron-dense granules. It has no internal thylakoid membrane system.

ACCEPT: Any unambiguous indication of C.

REJECT: A — chloroplast would show internal grana; B — mitochondrion shows cristae; D — nucleus shows chromatin/nucleolus.

Q3 *Plant-only structures and functions* **3 marks**

Award 1 mark per correct structure + function pair. Any three of:

• **Cell wall** — structural support / prevents excess water uptake / made of cellulose (1)

• **Middle lamella** — cements adjacent cells together / made of calcium pectate / pectin (1)

• **Plasmodesmata** — allow transport of substances / communication between adjacent cells (1)

• **Pits** — thin areas of cell wall allowing water movement between cells (1)

• **Vacuole (large, permanent)** — contains cell sap / maintains turgor pressure (1)

• **Tonoplast** — controls movement of substances into/out of the vacuole (1)

• **Chloroplast** — site of photosynthesis (1)

• **Amyloplast** — stores starch grains (1)

ACCEPT: any clearly equivalent wording for function.

REJECT: vague answers such as "gives structure" without specifying what or how.

Q4 *Carbohydrate storage differences* **2 marks**

Award 1 mark each for any two of:

• Plant cells store **starch**; animal cells store **glycogen** (1)

• Plant cells store carbohydrate inside **amyloplasts** (membrane-bound plastids); animal cells store glycogen granules **in the cytoplasm** / not in a membrane-bound organelle (1)

• Starch is a polymer of α-glucose; glycogen is also a polymer of α-glucose but is **more highly branched** (1) — accept as a second difference if first difference already given

ACCEPT: "plastid" in place of amyloplast.

REJECT: answers that say plant cells store starch "in the cytoplasm" (it is stored in amyloplasts).

Q5 *Cell wall diagram — identify X and Y; function of Y* **3 marks**

(a)

• X = **Middle lamella** (1)

ACCEPT: pectin layer / calcium pectate layer

• Y = **Plasmodesmata** (1)

ACCEPT: plasmodesma (singular); cytoplasmic channels

REJECT: "pits" — pits are thin areas of wall, not the channels shown.

(b)

• Y (plasmodesmata) allow transport of water / small molecules / signals between adjacent cells (1)

ACCEPT: connect cytoplasm of neighbouring cells; allow cell-to-cell communication.

REJECT: "allow exchange" without specifying what is exchanged.

Q6 *TEM organelle identification and tonoplast description* **4 marks**

(a)

• Organelle 1: **Chloroplast** (1) | Evidence: grana / stacked thylakoid membranes visible as parallel electron-dense lines (1)

ACCEPT: double outer membrane visible; pale stroma present.

• Organelle 2: **Amyloplast** (1) | Evidence: large very electron-dense starch granules inside / no internal membrane system (1)

ACCEPT: double outer membrane with dark irregular granules filling interior.

(b)

• The tonoplast appears as a **single, electron-dense membrane line** (1)

• It **surrounds / borders the large central vacuole** (1)

ACCEPT: located between the cytoplasm and the vacuole.

REJECT: "double membrane" — the tonoplast is a single membrane.

Q7 *Data table: identify polysaccharides P and Q; explain microfibril formation* **5 marks**

(a)

• P = **Cellulose** (1)

• Q = **Starch** (1)

ACCEPT: amylose specifically for Q.

(b) Award 1 mark each for any three of:

• Polysaccharide P (cellulose) has **straight / unbranched chains**, so parallel chains can lie close together (1)

• Parallel chains allow **hydrogen bonds to form between –OH groups** on adjacent chains (1)

• Many hydrogen bonds collectively hold chains together as **microfibrils** / give tensile strength (1)

• Polysaccharide Q has a **helical conformation** (from table), so chains cannot run parallel / cannot pack closely (1)

• Therefore Q cannot form inter-chain hydrogen bonds in the same way (1)

NOTE: Candidates do not need to use the term "microfibril" if the mechanism is correct.

Q8 *Extended writing — cellulose structure and function (AO1/AO2)* **6 marks**

Award 1 mark each for any **six** of the following (max 6):

• Cellulose is made of **β-glucose** monomers (1)

• β-glucose has its –OH on C1 pointing opposite to $–CH_2OH$ (opposite orientation to α-glucose) (1)

• Adjacent β-glucose monomers joined by **1,4-glycosidic bonds** via condensation reactions (1)

• Every alternate β-glucose **rotated 180°** to allow bond formation (1)

• This produces a **long, straight, unbranched chain** (1)

• Straight chains can lie **parallel** to each other (1)

• **Hydrogen bonds** form between –OH groups of parallel adjacent chains (1)

• Large numbers of hydrogen bonds — collectively strong despite individual weakness (1)

• Multiple parallel chains form a **cellulose microfibril** (1)

• Microfibrils arranged in **layers at different angles** in the cell wall (1)

• Gives the cell wall **high tensile strength** / resistance to stretching in multiple directions (1)

NOTE: Full-mark answers should follow the sequence: monomer → chain shape (why) → parallel chains → H-bonds → microfibril → cell wall strength. Answers jumping to "H-bonds give strength" without explaining why chains can form H-bonds (straight chain) should be capped at 4 marks.

Q9 *Synoptic — evaluate claim about cell wall composition* **3 marks**

Award 1 mark each for any three of:

• The statement is **incorrect** — both types of cell wall are rigid, but they have different compositions (1)

• Plant cell walls are made of **cellulose** (a polysaccharide of β-glucose) (1)

• Bacterial cell walls are made of **murein / peptidoglycan** (a glycoprotein / polysaccharide–peptide complex) (1)

• They are therefore **chemically different** despite both being rigid (1)

ACCEPT: reference to antibiotics (e.g. penicillin) targeting murein but not plant walls as an additional credit point.

NOTE: The student's error is assuming that the same function (rigidity) requires the same structure. Credit any answer that identifies this logical flaw.

End of mark scheme · 4A: The Cell Wall · Total: 25 marks · Edexcel IAL Biology Unit 2, Topic 4 · Specification refs 4.1(i), 4.1(ii), 4.2, 4.3